Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Core Pure 2 – Polar Coordinates

Medium & Hard • 25 Questions

Scan for free resources

Section A – Medium (Q1–15)

1. Convert the Cartesian point $(1, \sqrt{3})$ to polar form. **[3]**

2. Convert the polar point $\left(2, \frac{\pi}{4}\right)$ to Cartesian form. **[3]**

3. Sketch the curve $r = 2$. **[2]**

4. Sketch the curve $r = 2\cos\theta$. **[3]**

5. Sketch the curve $r = 1 + \cos\theta$ (cardioid). **[3]**

6. Find the Cartesian equation of $r = 4\cos\theta$. **[3]**

7. Find the area enclosed by $r = 2a\cos\theta$. **[4]**

8. Find $\dfrac{dy}{dx}$ in terms of r and θ for a polar curve. **[3]**

9. Find the points where the tangent is parallel to the initial line for $r = 1 + \sin\theta$. **[4]**

10. Convert $x^2 + y^2 = 4x$ to polar form. **[3]**

11. Find the area of the region enclosed by $r = a(1 + \cos\theta)$. **[4]**

12. Sketch $r = a\theta$ (Archimedean spiral) for $0 \le \theta \le 2\pi$. **[3]**

13. Find the polar equation of the circle centre origin radius a. **[2]**

14. Find the polar equation of the line $x = 2$. **[3]**

15. Find the maximum value of r for $r = 3 + 2\cos\theta$. **[2]**

Section B – Hard (Q16–25)

16. Find the area common to $r = 2\cos\theta$ and $r = 2\sin\theta$. **[6]**

17. Find the equation of the tangent to $r = 2a\cos\theta$ at $\theta = \frac{\pi}{3}$. **[5]**

18. Sketch $r^2 = a^2\cos 2\theta$ (lemniscate) and find the area of one loop. **[6]**

19. Find the points of intersection of $r = 1 + \cos\theta$ and $r = 1 - \cos\theta$. **[4]**

20. For $r = a(1 + \cos\theta)$, find the length of the arc from $\theta = 0$ to $\theta = \pi$. **[6]**

21. Convert $r = 2\sec\theta$ to Cartesian form and identify the curve. **[3]**

22. Find $\dfrac{dr}{d\theta}$ and the tangent of the angle ψ between the radius and the tangent for $r = a\theta$. **[5]**

23. Show that the area of a sector of a polar curve is $\frac{1}{2}\int r^2\,\mathrm{d}\theta$. **[3]**

24. Find the Cartesian equation of $r = 2a\sin\theta$. **[3]**

25. Sketch $r = 2 + 2\cos\theta$ and find the area it encloses. **[5]**

End of Worksheet • *Show all working clearly.*